

Armed Overdose

Section 1: Case Summary

Scenario Title:	Armed Overdose
Keywords:	Safety, Opioids, Overdose, Police, Violence
Brief Description of Case:	In this case a 44 y/o M is brought in via EMS after receiving 0.4mg of naloxone for what is suspected to be an opioid overdose. He remains GCS 7 upon arrival in the resuscitation bay. The team will need to work through the differential for altered LoC and will find drug paraphernalia and a loaded weapon on the patient upon inspection. The case will end with successful treatment and consultation with local police with regard to weapon and contraband protocols.

Goals and Objectives	
Educational Goal:	Safely manage a patient with a mixed overdose in the context of maintaining situation awareness and departmental guidelines around a patient found to be armed.
Objectives: (Medical and CRM)	CRM 1. Establish clear leadership 2. Delegate roles in the resuscitation of an acutely unwell patient 3. Communicate clearly amongst members of the team and employ closed loop communication Medical 1. Establish a differential diagnosis including sedative hypnotic and/or opioid toxidrome and trauma 2. Treat a patient with a mixed overdose syndrome 3. Effectively deal with a loaded weapon and drug paraphernalia using local protocols (e.g. RCMP or local police service)
EPAs Assessed:	N/A

Learners, Setting and Personnel	
Target Learners:	☐ Junior Learners ☒ Senior Learners ☒ Staff
	☒ Physicians ☒ Nurses ☒ RTs ☒ Inter-professional
	☐ Other Learners:
Location:	☐ Sim Lab ☒ In Situ ☐ Other:
Recommended Number of Facilitators:	Instructors: 1
	Sim Actors: 0
	Sim Techs: 1

Scenario Development	
Date of Development:	April 7, 2021
Scenario Developer(s):	Drs. Brad Stebner and Jared Baylis
Affiliations/Institutions(s):	UBC
Contact E-mail:	
Last Revision Date:	
Revised By:	
Version Number:	



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Unknown Male			Age: 40	Gender: M	Weight: 70kg
Presenting complaint: Altered LoC					
Temp: 35	HR: 80	BP: 100/60	RR: 12	O ₂ Sat: 96% r/a	FiO ₂ : 100% NRB
Cap glucose: 5.2 mmol/L			GCS: 7 (E1 V2 M4)		
Triage note: EMS notes state this patient was found downtown unresponsive. He was given one dose of 0.4mg IM naloxone and brought into the ED. Cap glucose 5.2 mmol/L. NRB at 15L/min applied. They also found a pool of blood around the patient and placed some gauze onto a bleeding scalp laceration.					
Allergies: Unknown					
Past Medical History: Unknown			Current Medications: Unknown		

Section 2B: Extra Patient Information

A. Further History	
<i>No further patient history is known. There is no available information in the EMR.</i>	
B. Physical Exam	
Cardio: Normal	Neuro: Low GCS, PERL 4mm
Resp: Normal	Head & Neck: Scalp laceration and small hematoma to left occiput
Abdo: Normal	MSK/skin: Track marks right ACF, has a loaded pistol and several baggies of "crystals" in pocket.
Other:	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Monitors IV access Crash cart
C. Required Medications
Naloxone Crystalloid
D. Moulage
Track marks right ACF Street clothes with loaded toy pistol tucked into pants and two small baggies of “crystals” in pockets Single 4x4 gauze soaked with ‘blood’
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> None aside from GCS 7 (E1, V2, M4)

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
RN	If patient not undressed for full exam within a few minutes, bedside nurse to prompt “we haven’t exposed the patient yet for examination”.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 80 BP: 100/60 RR: 12 O ₂ SAT: 96% T: 35°C GCS: 7		<u>Expected Learner Actions</u> ☐ Apply monitors ☐ Establish IV access ☐ Apply defibrillator pads ☐ Expose and examine patient ☐ Order appropriate labs and ECG ☐ Consider repeat naloxone dosing (although resp rate argues against this) ☐ Consider IV fluids for hypotension and point of care ultrasound for differential diagnosis ☐ Voice shared mental model with team including potential causes of altered level of consciousness ☐ Involve RCMP (police) once weapon located	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - naloxone-> minimal effect <u>Triggers</u> <i>For progression to next state</i> - after action items complete or 5 minutes → Phase 2
2. Persistent ALoC HR: 85 BP: 105/70 RR: 12 O ₂ SAT: 89% GCS: 5 (E1V1M4) No longer moaning to pain		<u>Expected Learner Actions</u> ☐ Consider CT head when not responding as expected ☐ Perform secondary survey to assess for other etiologies ☐ Consider pRBC transfusion when hypotension persists in context of scalp laceration	<u>Modifiers</u> - Repeat dosing of naloxone -> Minimal effect - Persistent bleeding from scalp laceration - Head stapled or sutured -> Bleeding resolves <u>Triggers</u>



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		☐ Consider discussion with poison control ☐ Liaise with RCMP (police) and charge nurse with regard to ED safety protocols ☐ Handle weapon +/- contraband according to local protocol	- Completion of action items → Phase 3	
3. Intubation HR: 90 BP: 95/60 RR: 10 O ₂ SAT: 89% -> 99% with intubation GCS: 3T		<u>Expected Learner Actions</u> ☐ Perform endotracheal intubation with appropriate analgesia/sedation/paralytics and PPE ☐ Consider peri-intubation IV crystalloid bolus + push dose pressor such as phenylephrine ☐ Post intubation OG/NG tube ☐ Post intubation CXR ☐ Post intubation sedation + analgesia ☐ Consult ICU and order CT head if not already done	<u>Modifiers</u> - Intubation → ΔO ₂ SAT to 99% <u>Triggers</u> - Completion of action items -> Phase 4	
4. Admission		<u>Expected Learner Actions</u> ☐ Admit to ICU for mixed opioid / sedative hypnotic overdose	END Scenario	

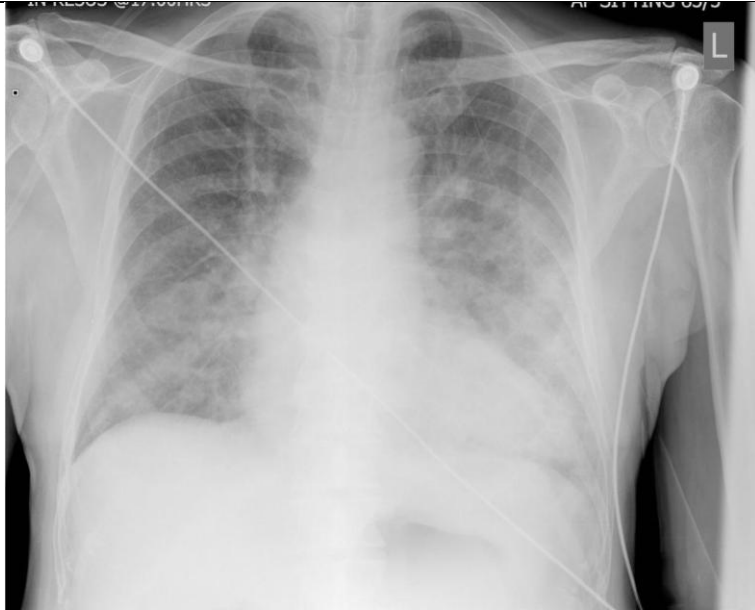
Simulation Scenario Template

Appendix A: Laboratory Results

<u>CBC</u> WBC 10.2 Hgb 145 Plt 376 <u>Lytes</u> Na 137 K 3.6 Cl 106 HCO ₃ 21 AG 10 Urea 5.7 Cr 75 Glucose 6.1 <u>Extended Lytes</u> Ca 2.12 Mg 0.73 PO ₄ 1.18 Albumin 42 TSH 2.37 <u>VBG</u> pH 7.29 pCO ₂ 38 pO ₂ 97 HCO ₃ 21 Lactate 3.3	<u>Cardiac/Coags</u> INR 1.0 aPTT 24 <u>Biliary</u> AST 29 ALT 22 GGT 12 ALP 70 Bili 12 Lipase 27 <u>Tox</u> EtOH <2 ASA <0.3 Tylenol <66
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Simulation Scenario Template

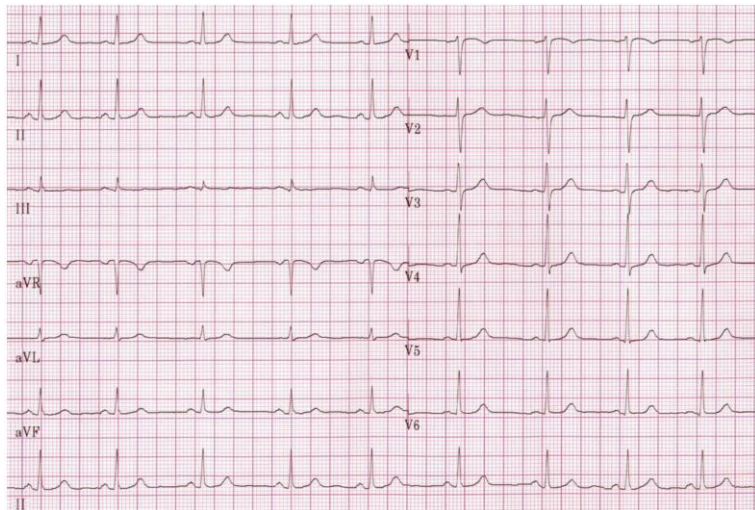
Appendix B: ECGs, X-rays, Ultrasounds and Pictures



Case courtesy of Dr Jeremy Jones, Radiopaedia.org, rID: 6463



Case courtesy of Dr David Cuete, Radiopaedia.org, rID: 23768



https://en.ecgpedia.org/index.php?title=Normal_Tracing

Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

- 1) What is the differential diagnosis for a patient with altered mental status?
- 2) What are your major safety concerns with a patient presenting with loaded weapons and drug paraphernalia?
- 3) What are your local protocols with regards to handling unknown illicit substances?
- 4) What is your local protocol for dealing with weapons found on patients?
- 5) What are your obligations as a medical professional in reporting incidents that involve weapons?

References

1. <https://emergencymedicinecases.com/opioid-misuse-overdose-withdrawal/>
2. <https://www.royalcollege.ca/rcsite/bioethics/cases/section-3/reporting-patients-harm-others-public-safety-e>
3. <https://pubmed.ncbi.nlm.nih.gov/16552279/>

